VOTE HER EKITI

# Technical Approach Proposal

*Gender Accountability Dashboard*

This proposal outlines the recommended architecture, implementation approach, delivery phases, and operational considerations for the Vote Her Ekiti Gender Accountability Dashboard, based on the supplied Product Requirements Document.

## 1. Project Understanding

The platform will provide a public-facing dashboard for monitoring gender representation and accountability in Ekiti State governance, alongside a secure administrative interface for authorised Women Too Africa Foundation (WTA) staff. The public website will display current appointment data, accountability scores, pledge and Charter statuses, historical trends, bulletins, and media resources. The admin interface will allow staff to maintain this information without routine developer intervention.

## 2. Proposed Technology Stack

| Layer | Proposed choice and purpose |
|---|---|
| **Frontend and application framework** | Next.js with TypeScript — for page rendering, routing, server-side logic, and a maintainable codebase. |
| **UI and styling** | Tailwind CSS and shadcn/ui — for responsive public pages, tables, forms, dialogs, and admin components. |
| **Database** | PostgreSQL — a relational database well suited to structured records and links between appointments, pledges, historical data, users, and audit logs. |
| **Database access** | Prisma ORM — to define database models and query PostgreSQL through TypeScript, with migrations to track schema changes. |
| **Charts** | Recharts — for gender-composition charts, historical comparisons, and accountability score breakdowns. |
| **Authentication** | An established authentication solution compatible with the chosen Next.js architecture — with secure sessions, password reset, and role-based permissions. Authentication will not be implemented from scratch. |
| **File storage** | Cloudinary or Amazon S3 — for PDFs, appointee photos, bulletin files, pledge documents, and the research report. |
| **Email** | Resend or an equivalent provider — for password- |

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| | reset messages and, if included in scope, bulletin notifications. |
| **Analytics and monitoring** | Plausible Analytics for visitor metrics and Sentry or an equivalent service for application error monitoring. |
| **Hosting** | Vercel for the Next.js application, plus managed PostgreSQL and external file storage. The final provider choices will be confirmed after checking cost, access, and deployment requirements. |

## 3. Application Architecture

- Public website: responsive pages for the homepage, Cabinet Tracker, Accountability Index, Pledge Archive, Charter, Historical Data, Bulletins, About, and Media Centre.
- Admin application: protected routes for overview, appointments, pledges, Charter statuses, bulletins, historical data, media, and settings.
- Server-side application logic: validates input, applies permissions, calculates metrics and scores, and exposes only the data required by each page.
- PostgreSQL database: stores structured records and their relationships. Uploaded PDF and image files will be stored in the selected file-storage service, with their references recorded in the database.
- Deployment environments: separate development and production configuration, with secrets held in environment variables and excluded from version control.

## 4. Core Data Models

- **Appointments:** Appointee name, position, gender, appointment date, category, district, status, source, photo reference, and timestamps.
- **Pledges and pledge history:** Official, position, signed date, summary, signed document, current status, status notes, and a record of each status change.
- **Charter demands:** Fixed demand number, category, approved demand text, response status, response date, and response notes.
- **Bulletins:** Edition, title, month/year, summary, PDF reference, publication schedule, and published/archive status.
- **Historical records:** Administration, years in office, total appointments, female appointments, and any additional fields required by the supplied research dataset.
- **Media coverage:** Publication, article title, date, summary, outbound link, and optional associated resources.
- **Users and roles:** Authorised users, secure authentication identifiers, and Administrator or Editor permissions.
- **Audit logs and settings:** Who changed what and when; scoring weights, target threshold, verdict labels, and configurable site metadata.

## 5. Key Implementation Decisions

- **Derived metrics:** Appointment counts, percentages, gaps to target, charts, and the overall index should be calculated from validated database records and approved scoring rules rather than maintained as unrelated hardcoded values.
- **Scoring methodology:** The four scoring dimensions and default weights will be configurable. Before implementation, WTA must approve the formula for each dimension, rounding rules, treatment of missing data, and whether historical scores are preserved or recalculated when the methodology changes.
- **Appointment scope:** The project team must confirm which appointment categories count towards the headline 35% State Executive Council target. Other categories can remain searchable in the tracker without necessarily being included in that calculation.
- **CSV operations:** CSV imports should validate columns and values, identify duplicate or invalid rows, and show a preview/error report before committing changes. Exports should respect the selected dataset and include appropriate column headers.
- **File uploads:** PDF and image uploads will have file-type and size validation. Files will be stored outside the relational database, with URLs or storage identifiers maintained in PostgreSQL.
- **Auditability:** Administrative changes, particularly status changes and scoring-setting updates, should be logged. Appointment records should be deactivated rather than permanently deleted through normal admin workflows.
- **Public transparency:** Every important metric should link to its underlying records or methodology. Sections without data should show a clear, approved explanation rather than a blank page or misleading zero.

## 6. Security, Performance and Accessibility

- Enforce HTTPS and protect all admin routes and server-side operations with role-based authorisation.
- Use a proven authentication library, secure password hashing/session handling, password-reset flows, and a 30-minute inactivity timeout where supported by the selected solution.
- Apply input validation, rate limiting to login endpoints, and protection against common web vulnerabilities including XSS, CSRF, and injection attacks.
- Keep credentials and API keys in environment variables; commit only a safe .env.example file.
- Configure automated daily database backups with at least 14 days of retention, and document restoration procedures.
- Optimise images and scripts, defer non-critical charts where appropriate, and test the homepage against the 3-second simulated 3G target and Lighthouse mobile score of 80 or above.
- Build responsive layouts for 320px-wide screens and test on a real low-end Android device.
- Target WCAG 2.1 AA: keyboard navigation, descriptive alt text, accessible chart alternatives, readable mobile typography, and status labels that do not rely on colour alone.
- Configure basic error monitoring and privacy-conscious visitor analytics.

## 7. Phased Delivery Plan

1. **Phase 0 — Scoping and setup:** Confirm formulas, appointment scope, content/data readiness, wireframes, database design, repository, environment setup, and acceptance criteria.

2. **Phase 1 — MVP:** Homepage with current score and headline metrics; Cabinet Tracker with core filters and CSV export/import as agreed; initial admin authentication and appointment management; bulletin upload/management; responsive design and first deployment.
3. **Phase 2 — Accountability modules:** Detailed Index breakdown; Pledge Archive and status history; Charter Tracker; Historical Data charts and tables; bulletin archive; scorecard social-sharing image.
4. **Phase 3 — Completion and optimisation:** About page; Media Centre; accessibility and performance audit; optional Yoruba labels if translations are supplied and feature is confirmed in scope; final security and operational checks.
5. **Handover:** Source-code handover, README and setup/deployment documentation, environment-variable guide, admin training session, backup/restore notes, and agreed support terms.

Schedule note: The MVP target at the end of October 2026 is tight given the breadth of the required public and admin functionality. A serious effort will be made to meet the milestones, but delivery depends on prompt access to approved data and content, timely client feedback, and early agreement on the essential MVP features. If dependencies are delayed, the parties should agree in writing on revised priorities or dates.

## 8. Client Responsibilities and Dependencies

- Provide accurate, approved appointment and historical datasets in an agreed spreadsheet/CSV format.
- Provide the ratified Charter, signed pledges, research report, bulletins, media links, press-kit assets, and required summaries.
- Provide official branding assets, project copy, contact information, and any translations included in scope.
- Approve the scoring formulas, target calculation rules, data treatment, and publication workflow.
- Nominate a primary contact who can consolidate feedback and make decisions within the agreed review window.
- Own or control the domain and third-party service accounts, and pay all related charges directly. These costs are excluded from the developer's fee.
- Review each milestone and report defects against the agreed acceptance criteria promptly.

## 9. Assumptions, Exclusions and Change Control

- The proposal is based on the requirements in the supplied PRD. Features not listed or materially expanded after approval will be treated as change requests.
- Content writing, independent research, data verification, translation, and preparation of missing source datasets are client responsibilities unless separately agreed.
- Domain registration, hosting, database, file storage, email, analytics, monitoring, and other third-party subscription or usage charges are payable by WTA and are not included in the development fee.
- Any ongoing maintenance or post-handover bug-fix support must be defined in a separate written agreement. The 30-day free support period stated in the PRD is not accepted as written.
- The exact milestone dates should be confirmed after Phase 0 and after assessing the readiness of data, content, and access to required services.

## 10. Handover Deliverables

- Private GitHub repository owned by or shared with WTA.

- Application source code and database schema/migrations.
- README with local setup, environment variables, deployment, and admin instructions.
- A .env.example file without secrets.
- Admin training session and agreed documentation.
- Production deployment and handover of relevant account access, with WTA retaining ownership/control of its domain and paid services.

## 11. Approval and Next Steps

The proposed approach is intended to provide a maintainable, secure, and data-driven platform that can be extended or replicated for similar initiatives. Implementation will begin after WTA confirms the scoring methodology, appointment-counting rules, data readiness, phase acceptance criteria, service ownership, and the revised post-launch support arrangement.